

AMANGELDI, Kazakhstan

WMO: 353610

Lat: 50.133N

Lon: 65.233E

Elev: 142

StdP: 99.63

Time Zone: 6.00 (E06)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-31.0	-27.4	-33.6	0.2	-30.7	-30.5	0.2	-27.2	20.2	-4.9	18.5	-14.4	4.4	230	0.656

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.3	35.0	19.7	33.1	19.0	31.2	18.2	21.4	31.1	20.2	30.2	19.3	29.0	4.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	12.9	26.6	16.6	12.0	25.3	15.4	11.1	24.1	62.6	31.1	58.8	30.4	55.5	28.9	28.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.1	14.3	11.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.6	-14.5	-14.8	-6.8	7.6	14.6	20.8	23.2	20.3	13.8	6.0	-4.2	-11.7
	DBStd	14.75	7.25	6.56	6.70	5.61	4.60	4.46	3.30	3.79	4.41	4.51	5.38	6.77
	HDD10.0	3348	760	693	521	109	13	0	0	0	14	138	427	672
	HDD18.3	5364	1018	927	779	322	130	26	4	23	146	381	677	930
	CDD10.0	1393	0	0	0	38	156	325	410	321	128	16	0	0
	CDD18.3	367	0	0	0	0	15	101	156	86	10	0	0	0
	CDH23.3	4593	0	0	0	18	219	1336	1828	1030	158	4	0	0
	CDH26.7	1953	0	0	0	3	53	609	826	420	41	1	0	0
(14) Wind	WSAvg	4.8	6.1	5.7	5.9	5.0	4.5	4.2	4.0	4.0	4.0	4.3	4.8	4.8
(15) Precipitation	PrecAvg	435	39	36	39	30	34	33	48	31	21	33	44	46
	PrecMax	601	77	98	82	56	82	96	91	77	54	92	91	85
	PrecMin	311	10	5	2	2	8	8	15	3	1	4	14	17
	PrecStd	81	19	19	20	13	17	22	22	20	13	18	19	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	0.5	12.0	25.8	30.7	37.4	37.5	35.7	30.8	22.4	8.8	1.8
		MCWB	0.1	-0.6	6.1	12.9	17.2	20.6	20.3	18.8	17.9	13.1	4.5	0.5
	2%	DB	-0.8	-1.1	6.9	22.4	28.2	35.0	35.2	33.0	27.5	19.0	5.6	0.2
		MCWB	-1.4	-1.9	3.5	12.4	17.4	20.0	19.8	17.9	15.1	11.0	3.7	-0.7
	5%	DB	-2.7	-2.6	3.9	19.7	25.9	32.8	33.2	31.0	25.3	16.3	3.8	-0.8
		MCWB	-3.3	-3.3	1.7	11.7	15.9	19.1	19.5	17.7	14.3	9.5	2.2	-1.7
	10%	DB	-4.5	-4.1	1.8	17.1	23.7	30.6	31.2	29.1	22.9	13.9	2.2	-2.2
		MCWB	-5.0	-4.8	0.2	10.4	14.6	18.1	18.6	16.9	13.4	8.2	1.0	-2.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.5	-0.4	6.9	14.5	21.1	23.9	23.0	21.2	19.0	13.7	5.9	0.7
		MCDB	0.8	0.2	10.4	22.3	27.2	33.6	33.5	30.3	28.5	20.6	7.6	1.5
	2%	WB	-1.4	-1.8	4.0	13.3	18.2	21.3	21.4	19.8	16.9	11.7	3.9	-0.5
		MCDB	-0.9	-1.2	6.7	20.7	26.2	32.5	31.0	30.0	24.7	17.7	5.3	0.2
	5%	WB	-3.3	-3.3	1.8	12.1	16.5	19.9	20.4	18.5	15.1	10.0	2.3	-1.5
		MCDB	-2.8	-2.6	3.8	18.6	23.9	30.4	30.1	28.4	23.1	15.6	3.7	-0.8
	10%	WB	-5.0	-4.7	0.4	10.8	15.3	18.7	19.6	17.6	13.7	8.5	1.1	-2.9
		MCDB	-4.5	-4.0	1.8	17.0	22.2	29.0	29.1	27.0	21.5	13.5	2.2	-2.2
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	8.9	9.2	11.0	12.9	14.2	13.3	13.7	13.3	10.8	7.0	7.5
		MCDBR	9.2	8.7	9.8	15.3	16.7	17.0	16.1	17.1	17.2	15.6	8.1	8.0
		MCWBR	8.8	8.1	7.1	7.3	8.0	7.5	5.8	6.9	8.3	9.4	6.6	7.5
	5% WB	MCDBR	8.9	8.3	8.8	13.9	14.7	16.0	14.4	15.2	15.4	14.2	7.5	8.0
		MCWBR	8.6	8.0	6.7	7.5	7.8	8.0	6.5	7.0	8.2	9.1	6.4	7.6
(40) Clear-Sky Solar Irradiance	taub		0.306	0.332	0.393	0.423	0.417	0.415	0.420	0.413	0.362	0.350	0.340	0.310
	taud		2.051	2.000	2.003	2.096	2.189	2.256	2.257	2.255	2.346	2.295	2.184	2.089
	Ebn at Noon		687	770	789	816	841	845	833	819	827	759	649	618
	Edh at Noon		86	118	142	145	138	131	129	123	101	88	77	73
(44) All-Sky Solar Radiation	RadAvg		1.44	2.61	3.83	4.95	6.40	6.78	6.36	5.58	4.23	2.41	1.15	0.95
	RadStd		0.17	0.21	0.41	0.43	0.37	0.38	0.40	0.41	0.26	0.27	0.12	0.14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page